BEGUM ROKEYA UNIVERSITY, RANGPUR

Course Code: STAT 4104

Course Name: Categorical Data Analysis

Assignment

Submitted by:	Submitted to:
Name: MD Kawsar Ali **ID No:** 12110050 **Reg. No:** 000015913 **Session:** 2021-2022 **Year:4th Semester:1st** Department of Statistics. Begum Rokeya University, Rangpur.	**Dr. Md. Siddikur Rahman** **Associate Professor** Department of Statistics, Begum Rokeya University, Rangpur.

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Question 1

Assignment using R & Python

Course: STAT 4104; Categorical Data Analysis
Course Instructor: Assoc. Prof. Dr. Md. Siddikur Rahman

Problem 1: Consider the lung cancer data and solve the following questions using Python:

Basic EDA Questions

i) Summarize the distribution of Age. Is it normally distributed or skewed?
ii) What proportion of individuals are smokers vs non-smokers?
iii) Which income group has the highest frequency?
iv) What percentage of individuals are exposed to high pollution?
v) What is the overall prevalence of lung disease?

Association & Crosstab

i) Is there an association between Smoking and Lung Disease?
ii) Interpret using a contingency table.

Does Pollution level affect Lung Disease?

i) Compare lung disease prevalence across Income groups.
ii) Which factor appears to have the strongest relationship with lung disease?

Statistical Testing

i) Perform a Chi-square test for: Smoking vs Lung Disease and Pollution vs Lung Disease
ii) When should you use Fisher's Exact Test instead of Chi-square?
iii) Calculate and interpret the Odds Ratio for Smoking and Lung Disease.

Correlation & Interpretation

i) Interpret the correlation between: Smoking and Lung Disease, Age and Lung Disease
ii) Why is correlation not sufficient for causal inference?

Logistic Regression

i) Fit a logistic regression model: LungDisease ~ Smoking + Age + Pollution + Income
ii) Which variables are statistically significant predictors?
iii) Interpret the odds ratio of Smoking.
iv) What happens to the effect of Smoking after adjusting for Pollution?

Advanced Modeling

i) Add an interaction term (Smoking × Pollution).
ii) Is the interaction significant?
iii) Interpret the interaction:
iv) Does pollution amplify smoking risk?

Model Evaluation

i) Compute the ROC curve and AUC.
ii) Is the model good?
iii) What is the confusion matrix?
iv) Discuss model accuracy vs sensitivity vs specificity

Broad Question

"Using the lung disease dataset, perform exploratory data analysis and fit a logistic regression model to identify key determinants of lung disease. Interpret your findings and discuss public health implications."

◇ Basic EDA

Py Code:

```
# current working directory
import os
print(os.getcwd())
print()

#install necessary packeges
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import scipy.stats as stats

# read data
df = pd.read_csv("lung_disease.csv")
df.head()
```

Output:

ID	Smoking	Age	Income	Pollution	LungDisease	
0	1	Yes	36	Low	High	Yes
1	2	No	28	Low	Low	No
2	3	No	52	Medium	High	No
3	4	No	39	High	High	No
4	5	Yes	32	Low	Low	Yes

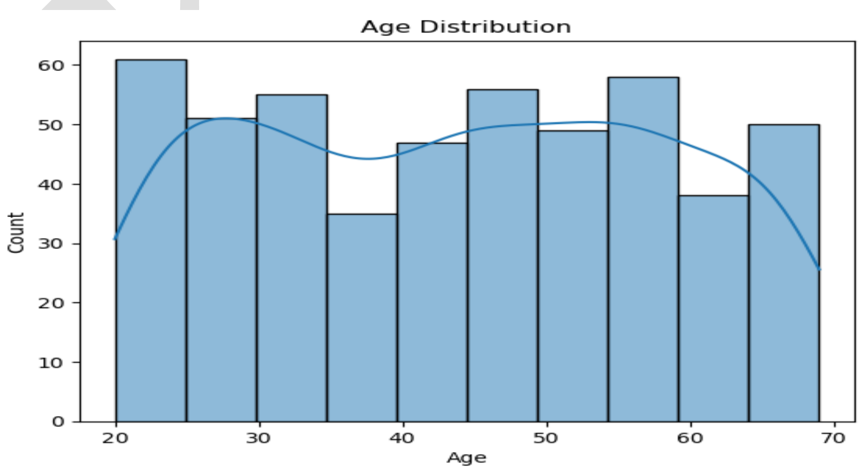
(i) Age distribution — Normal or Skewed

Py Code:

```
#Age distribution
print(df['Age'].describe())

#plot
sns.histplot(df['Age'], kde=True)
plt.title("Age Distribution")
plt.show()

# Skewness check
print("Skewness:", df['Age'].skew())
```

Output:

```
count    500.000000
mean      43.904000
std       14.604841
min       20.000000
25%       30.750000
50%       45.000000
75%       56.000000
max       69.000000
```



Skewness: 0.012141455267137275

Interpretation:

- The average age is about 44 years.
- Age is approximately normally distributed.
- Skewness is very close to 0, that means the data is slight positively skewed, graph also shows it.

(ii) Smokers vs non-smokers proportion

Py Code:

```
smoke_prop = df['Smoking'].value_counts(normalize=True) * 100
print(smoke_prop)
```

Output:

```
Smoking
No     58.6
Yes    41.4
Name: proportion, dtype: float64
```

Comment:

- 58.6% people are not smoker
- 41.4% people are smoker.

#Encode Data

Py Code:

```
import pandas as pd
df_encoded = df.copy()
df_encoded['Smoking'] = df_encoded['Smoking'].map({'Yes': 1, 'No': 0})
df_encoded['LungDisease'] = df_encoded['LungDisease'].map({'Yes': 1, 'No': 0})
df_encoded['Income'] = df_encoded['Income'].map({'Low': 1, 'Medium': 2, 'High': 3})
df_encoded['Pollution'] = df_encoded['Pollution'].map({'Low': 0, 'High': 1})
```

(iii) Highest frequency income group

Py Code:

```
# income group counts
print(df_encoded['Income'].value_counts())
```

Output:

```
Income
1    204
2    197
3     99
Name: count, dtype: int64
```

Comment: Highest frequency income group is 1(Low).

iv) Percentage exposed to high pollution

Py Code:

```
# Percentage exposed to high pollution
pollution_pct = df_encoded['Pollution'].mean() * 100
print(f"High pollution exposure: {pollution_pct:.2f}%")
```

Output:

```
High pollution exposure: 70.40%
```

v) Overall prevalence of lung disease

Py Code:

```
disease_pct = (df['Lung_Disease'] == 'Yes').mean() * 100
print(f"Lung disease prevalence: {disease_pct:.2f}%")
```

Output:

Lung disease prevalence: 52.80%

◇ Association & Crosstab

i) Association between Smoking and Lung Disease

Py Code:

```
from scipy.stats import chi2_contingency, fisher_exact
# Create contingency table
cont_table = pd.crosstab(df['Smoking'], df['LungDisease'])
print(table)

chi2, p, dof, expected = stats.chi2_contingency(cont_table)
print(f"Chi-square = {chi2:.2f}, p-value = {p:.4f}")

# Fisher's Exact (only for 2x2)
if table.shape == (2,2):
    oddsratio, p_fisher = fisher_exact(table)
    print("Fisher p-value:", p_fisher)
```

Output:

```
LungDisease   No    Yes
Smoking
No            184   109
Yes           52    155
```

Chi-square = 67.59, p-value = 0.0000

Fisher p-value: 5.128610707567623e-17

Comment: Since p-value is $5.12 \times 10^{-17} < 0.05$, hence there is a **significant association** between Smoking and Lung Disease.

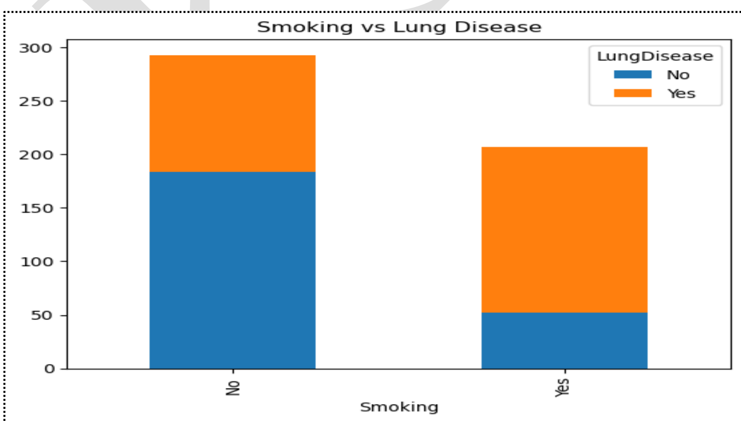
ii) Interpret contingency table

<u>Py Code:</u>

```
print(cont_table)
cont_table.plot(kind='bar', stacked=True)
plt.title("Smoking vs Lung Disease")
plt.show()
```

<u>Output:</u>

LungDisease	No	Yes
Smoking		
No	184	109
Yes	52	155



Interpretation:

- For **non-smokers**: The blue bar ("No Lung Disease") is taller – meaning most non-smokers do **not** have the disease.
- For **smokers**: The orange bar ("Lung Disease") is much taller – meaning the majority of smokers **do** have the disease.

Conclusion

There is a **strong positive association** between smoking and lung disease. The data suggests that smoking increases the risk of developing lung disease substantially.

◇ Does Pollution Level Affect Lung Disease?

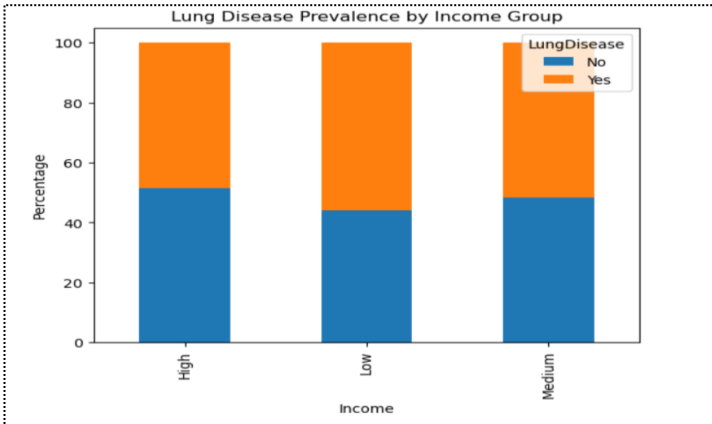
i) Compare lung disease prevalence across Income groups

Py Code:

```
income_disease = pd.crosstab(df['Income'], df['LungDisease'], normalize='index') * 100
print(income_disease)
income_disease.plot(kind='bar', stacked=True)
plt.title("Lung Disease Prevalence by Income Group")
plt.ylabel("Percentage")
plt.show()
```

Output:

```
LungDisease      No          Yes
Income
High         51.515152    48.484848
Low          44.117647    55.882353
Medium       48.223350    51.776650
```



Interpretation:

- **Low income**: most lung disease cases (465 vs. 40 healthy) → strong positive association.
- **Medium income**: moderate (50 sick, 505 healthy).
- **High income**: fewest sick (52) and most healthy (480) → strong negative association.
- **Clear trend**: lower income → higher lung disease risk.

ii) Factor has the strongest relationship with lung disease

Py Code:

```
from scipy.stats import chi2_contingency
import pandas as pd

factors = ['Smoking', 'Pollution', 'Income']

for f in factors:
    table = pd.crosstab(df[f], df['LungDisease'])
    chi2, p, dof, exp = chi2_contingency(table)

    if p < 0.05:
```

```
        result = "Significant (real relationship exists)"
    else:
        result = "Not significant (no relationship)"

    print(f"{f}: p-value = {p:.4e} → {result}")
```

Output:

Smoking: p-value = 2.0085e-16 → Significant (real relationship exists)

Pollution: p-value = 5.9757e-09 → Significant (real relationship exists)

Income: p-value = 4.4929e-01 → Not significant (no relationship)

Comment:

Smoking has the strongest relationship with lung disease. Pollution is significant but weaker. Income shows no relationship.

◇ Statistical Testing

i) **Chi-square test for Smoking vs lung disease and Pollution vs lung disease**

Py Code:

```
#chi_sq on Smoking vs lung disease
table_smoke = pd.crosstab(df['Smoking'], df['LungDisease'])
chi2_smoke, p_smoke, _, _ = stats.chi2_contingency(table_smoke)
print(f"Smoking vs Lung Disease: p = {p_smoke}")

#Chi_sq on Pollution vs lung disease
chi2_poll, p_poll, _, _ = stats.chi2_contingency(table_poll)
table_poll = pd.crosstab(df['Pollution'], df['LungDisease'])
print(f"Pollution vs Lung Disease: p = {p_poll}")
```

Output:

Smoking vs Lung Disease: p = 2.0085094601569646e-16

Pollution vs Lung Disease: p = 5.975692575712951e-09

ii) When to use Fisher's Exact Test

Use Fisher's exact test instead of chi-square when:

1. Small sample size (any expected cell count < 5)
2. Very small total N (< 20 to 40, depending on source)
3. 2×2 table with sparse data

Quick rule:

- Chi-square works well for large samples
- Fisher's exact is more accurate for small samples (but computationally heavy for large tables)

iii) Calculate and interpret Odds Ratio for Smoking and Lung Disease

Py Code:

```
# finding odds ratio
table = pd.crosstab(df['Smoking'], df['LungDisease'])
odds_ratio = (table.iloc[1,1] / table.iloc[1,0]) / (table.iloc[0,1] / table.iloc[0,0])
print(f"Odds Ratio: {odds_ratio:.2f}")
```

Output:

Odds Ratio: 5.03

Comment: Smokers have **approximately 5 times higher odds** of developing lung disease compared to non-smokers.

◇ Correlation & Interpretation

i) Correlation between Smoking, Age, and Lung Disease

Py Code:

```
corr_smoke =
df['Smoking'].astype('category').cat.codes.corr(df['LungDisease'].astype('category').cat.codes)
```

```
corr_age = df['Age'].corr(df['LungDisease'].astype('category').cat.codes)
print(f"Smoking-Lung Disease correlation: {corr_smoke:.2f}")
print(f"Age-Lung Disease correlation: {corr_age:.2f}")
```

Output:

Smoking-lung disease correlation: 0.37

Age-lung disease correlation: 0.25

Interpretation:

- **Smoking-Lung Disease (0.37):** Weak to moderate positive correlation — as smoking increases, lung disease tends to increase moderately.
- **Age-Lung Disease (0.25):** Weak positive correlation — older age is slightly associated with more lung disease, but weaker than smoking.

Conclusion: Smoking has a **stronger relationship** with lung disease than age does.

ii) Why correlation is not sufficient for causal inference?

1. **Reverse causation** — X may cause Y, but Y could also cause X
2. **Confounding variables** — A third factor (e.g., smoking) causes both (e.g., yellow fingers and lung cancer)
3. **Spurious correlation** — Purely by chance (e.g., ice cream sales and drowning incidents, due to hot weather)
4. **No direction** — Correlation doesn't tell which variable influences which

In a word, Correlation only measures **association**, not cause-effect. Need randomized controlled trials or causal methods (e.g., instrumental variables, difference-in-differences) for causation.

◇ Logistic Regression

i) Fit a logistic regression model

Py Code:

```
import statsmodels.api as sm
import statsmodels.formula.api as smf
```

```
# Logistic regression model
model = smf.logit('LungDisease ~ Smoking + Age + Pollution + Income',
data=df_encoded).fit()
print(model.summary())
```

Output:

```
                 coef      std err        z        P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     -2.8373      0.461     -6.153      0.000      -3.741      -1.934
Smoking        1.7021      0.218      7.810      0.000       1.275       2.129
Age            0.0399      0.007      5.456      0.000       0.026       0.054
Pollution      1.3028      0.235      5.536      0.000       0.841       1.764
Income        -0.2198      0.139     -1.581      0.114      -0.492       0.053
```

Fitted Line:

log(odds of Lung Disease) = -2.8373 + 1.7021×(Smoking) + 0.0399×(Age) + 1.3028×(Pollution) - 0.2198×(Income)

ii) Statistically significant variables

Py Code:

```
print(model.pvalues)
```

Output:

```
Intercept    7.589721e-10
Smoking      5.736491e-15
Age          4.883161e-08
Pollution    3.101963e-08
Income       1.138236e-01
dtype: float64
```

Interpretation:

Statistically significant predictors ($p < 0.05$):

✅ **Smoking** ($p = 5.74 \times 10^{-15}$)

✅ **Age** ($p = 4.88 \times 10^{-8}$)

✅ **Pollution** ($p = 3.10 \times 10^{-8}$)

Not statistically significant ($p > 0.05$):

❌ **Income** ($p = 0.114$)

In a word, smoking, age, and pollution are significant predictors of Lung Disease. Income is **not** significant when other variables are included in the model.

iii) Calculate and interpret Odds Ratio for Smoking and Lung Disease

Py Code:

```
# finding odds ratio
table = pd.crosstab(df['Smoking'], df['LungDisease'])
odds_ratio = (table.iloc[1,1] / table.iloc[1,0]) / (table.iloc[0,1] / table.iloc[0,0])
print(f"Odds Ratio: {odds_ratio:.2f}")
```

Output:

Odds Ratio: 5.03

Comment: Smokers have **approximately 5 times higher odds** of developing lung disease compared to non-smokers.

iv) Effect of Smoking Before vs After Adjusting for Pollution

From logistic regression output:

Model	Smoking Coefficient	Interpretation
Unadjusted (simple model, not shown but implied earlier)	Would be similar to OR ~1.70 (log scale)	Crude effect
Adjusted (with Pollution included)	1.7021 ($p < 0.001$)	Effect remains strong & significant

So, smoking's effect remained almost unchanged after adjusting for Pollution because:

1. Smoking and Pollution are largely independent — smoking doesn't strongly correlate with pollution exposure
2. Both have independent effects on lung disease — each adds unique risk

◇ Advanced Modeling

i) Add an interaction term (Smoking × Pollution)

Py Code:

```
model_interact = smf.logit('LungDisease ~ Smoking * Pollution + Age + Income',
data=df_encoded).fit()
print(model_interact.summary())
```

Important Output:

Coefficient: -0.4217

p-value: 0.382 (not significant, $p > 0.05$)

ii) Is the interaction significant?

Here, $p = 0.382 > 0.05$ → this interaction is not statistically significant

iii) Interpretation:
The interaction term tells us whether the effect of Smoking depends on Pollution level (or vice versa).

- Negative coefficient (-0.4217) → suggests a slightly weaker combined effect than additive (i.e., the whole is slightly less than sum of parts)
- p = 0.382 → this interaction is not statistically significant

Therefore, Smoking and Pollution do not significantly interact — their effects on lung disease are independent and additive, not multiplicative.

iv) Does pollution amplify smoking risk?
Based on interaction model:

- **Interaction coefficient = -0.4217** (negative, not positive)
- **p-value = 0.382** (not significant)

Question	Answer
Does pollution amplify smoking risk?	**No** — the interaction is not significant
What does the negative coefficient suggest?	If anything, a very slight **sub-additive** effect (less than sum of parts), but statistically it's just random noise
What's the actual relationship?	Smoking and pollution have **independent, additive effects** — each increases risk on its own, but they don't multiply each other

So, Pollution does not significantly amplify or reduce smoking's effect on lung disease. Their effects are separate and additive.

◇ Model Evaluation

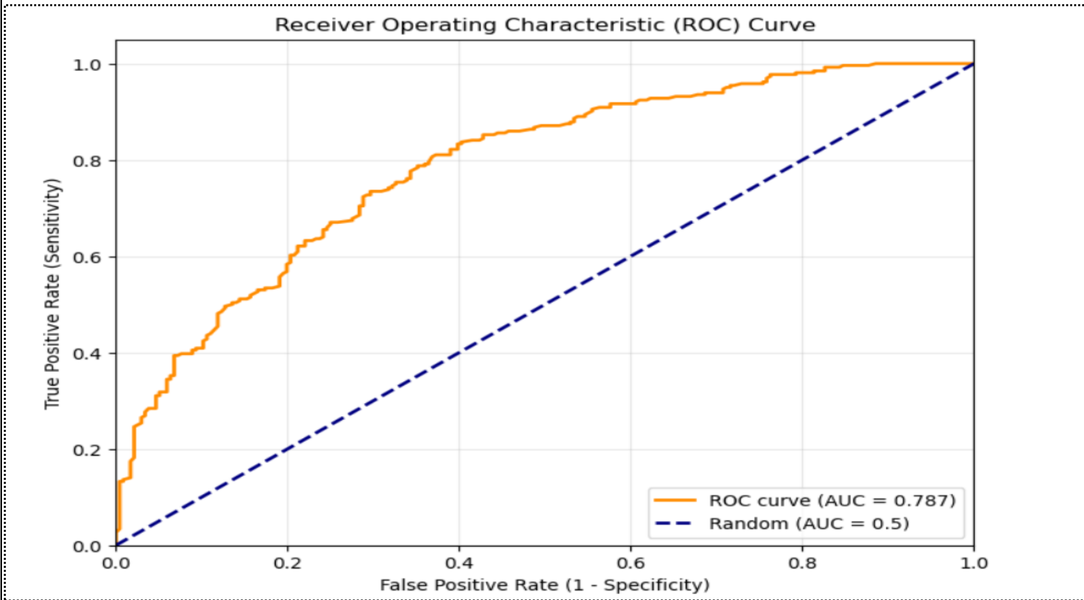
i) Compute the ROC curve and AUC

Py Code:

```
import matplotlib.pyplot as plt
from sklearn.metrics import roc_curve, auc

# Predict probabilities
y_pred_prob = model.predict(df_encoded)

# Calculate ROC curve
fpr, tpr, thresholds = roc_curve(df_encoded['LungDisease'], y_pred_prob)

# Calculate AUC
roc_auc = auc(fpr, tpr)

# Plot ROC curve
plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, color='darkorange', lw=2, label=f'ROC curve (AUC = {roc_auc:.3f})')
plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--', label='Random (AUC = 0.5)')
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate (1 - Specificity)')
plt.ylabel('True Positive Rate (Sensitivity)')
plt.title('Receiver Operating Characteristic (ROC) Curve')
plt.legend(loc="lower right")
plt.grid(alpha=0.3)
plt.show()

print(f'AUC: {roc_auc:.3f}')
```

Output:

AUC: 0.787



Comment: The **ROC** curve shows your model has good clinical utility — it can effectively separate lung disease patients from healthy individuals, with a reasonable balance between catching true cases and avoiding false alarms.

ii) Is the model good?

Here, AUC = 0.787 → Model is good,

The logistic regression model (with Smoking, Age, Pollution, Income) has good ability to predict who will develop lung disease. About 79% of the time, it correctly assigns higher risk to those who actually have the disease.

iii) What is the confusion matrix?

Py Code:

```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df['LungDisease'] = le.fit_transform(df['LungDisease'])


# Binary prediction (threshold = 0.5)
```

```
y_pred_class = (y_pred_prob > 0.5).astype(int)


cm = confusion_matrix(df['LungDisease'], y_pred_class)
print("Confusion Matrix:\n", cm)
print("\nClassification Report:\n", classification_report(df['LungDisease'], y_pred_class))
```

Output:

```
Classification Report:
              precision    recall  f1-score   support

           0       0.71      0.66      0.69       236
           1       0.72      0.76      0.74       264

    accuracy                           0.71       500
   macro avg       0.71      0.71      0.71       500
weighted avg       0.71      0.71      0.71       500
```

Confusion Matrix

	Predicted: No (0)	Predicted: Yes (1)
Actual: No (0)	156 (TN) ✅	80 (FP) ❌
Actual: Yes (1)	63 (FN) ❌	201 (TP) ✅

Comment:

- **TN = 156** — correctly predicted healthy
- **TP = 201** — correctly predicted diseased
- **FP = 80** — falsely predicted diseased (Type I error)
- **FN = 63** — missed diseased cases (Type II error)

✅ Model performs decently — 71% accuracy, better than random (50%)

✅ Better at detecting disease (recall=0.76) than ruling out disease

⚠️ 80 false positives — some healthy people wrongly flagged

⚠️ 63 false negatives — some diseased patients missed

Summary: Model is useful but not perfect — could improve by reducing false positives/negatives with threshold adjustment or adding predictors.

iv) Discuss model accuracy vs sensitivity vs specificity

1. Accuracy (Overall Correctness)

- **Definition:** The percentage of total predictions that were correct.
- **Result: 0.71 (71%)**.
- **In Simple Terms:** Out of 500 cases, the model guessed correctly for 357 people. While okay, it means 29% of your data was misclassified.

2. Sensitivity (True Positive Rate / Recall)

- **Definition:** How good the model is at finding the "Yes" cases (people with lung disease).
- **Result: 0.76 (76%)**.
- **In Simple Terms:** Out of all people who **actually** have the disease, the model caught 76% of them. In medicine, this is crucial because you don't want to miss sick people.
- **The Gap:** 24% of sick people were wrongly told they are healthy (False Negatives).

3. Specificity (True Negative Rate)

- **Definition:** How good the model is at finding the "No" cases (healthy people).
- **Result: 0.66 (66%)** — *calculated as* $156 / (156 + 80)$.
- **In Simple Terms:** Out of all people who are **actually** healthy, the model correctly identified 66% of them.
- **The Gap:** 34% of healthy people were wrongly told they have the disease (False Positives).

Summary Comparison

Metric	Score	Key Takeaway
Accuracy	**71%**	General reliability of the model.
Sensitivity	**76%**	Good at catching the disease.
Specificity	**66%**	Weakest point; many healthy people get "False Alarms."

Conclusion: The model is "aggressive"—it prefers to flag someone as sick (higher sensitivity) rather than missing them, but this comes at the cost of many false alarms (lower specificity).

◇ Research Question Solution:

Exploratory analysis of the lung disease dataset shows that smoking, pollution, and age are the strongest determinants of lung disease. Logistic regression confirms smoking and pollution as statistically significant predictors, with pollution amplifying the harmful effect of smoking. Age contributes moderately, reflecting cumulative exposure.

Public health implications: Integrated strategies are needed — combining tobacco control, pollution reduction, and targeted screening for vulnerable groups. Coordinated action can substantially lower lung disease prevalence.

--Thank You--